

Method 5: BERNOULLI'S EQUATION.

Lesson 23

If a 1st order differential equation is non-linear and of the form below, it is called Bernoulli's equation:

$$\frac{dy}{dx} + p(x)y = f(x)y^n \quad (n \neq 0, 1)$$

Standard Bernoulli eqn

Procedure for solving:

Step 1 $\{ \times y^{-n}$ both sides:

$$y^{-n} \frac{dy}{dx} + p(x)y^{1-n} = f(x)$$

Step 2 $\{ \times (1-n)$:

$$(1-n)y^{-n} \frac{dy}{dx} + (1-n)p(x)y^{1-n} = (1-n)f(x) \quad (*)$$

Step 3 $\{$ Substitute:

$$\text{Let } Z = y^{1-n} \rightarrow \frac{dZ}{dx} = (1-n)y^{-n} \frac{dy}{dx}$$

then in step 2, we have:

$$\frac{dZ}{dx} + (1-n)p(x)Z = (1-n)f(x) \quad \leftarrow \text{Linear form}$$

This is now linear d.e and can be solved using I.F-method.

Example 1

$$\text{Solve: } \frac{dy}{dx} + \frac{1}{x}y = xy^2$$

Solution

Step 1: $\{ \times y^{-2}$:

$$y^{-2} \frac{dy}{dx} + \frac{1}{x}y^{-1} = x$$

Step 2: $\{ \times (-1)$:

$$-1y^{-2} \frac{dy}{dx} - \frac{1}{x}y^{-1} = -x$$

Step 3: Let $Z = y^{-1} \xrightarrow{\frac{d}{dx}} \frac{dZ}{dx} = -1 \cdot y^{-2} \frac{dy}{dx}$

Substitute: $\frac{dz}{dx} - \frac{1}{x}z = -x$ ← (Linear) -2-

Apply I.F method:

$$p(x) = -\frac{1}{x}; \quad \int p(x) dx = \int -\frac{1}{x} dx = -\ln x = \ln(x^{-1}) = \ln\left(\frac{1}{x}\right)$$

$$I.F = e^{\ln(\frac{1}{x})} = \frac{1}{x}$$

$$\frac{1}{x} \left(\frac{dz}{dx} - \frac{1}{x}z \right) = -x \cdot \frac{1}{x}$$

$$\frac{d}{dx} \left(z \cdot \frac{1}{x} \right) = -1$$

$$z \cdot \frac{1}{x} = \int -1 dx = -x + C$$

$$\frac{1}{y} = y^{-1} = z = -x^2 + Cx$$

$$y = \frac{1}{Cx - x^2}$$

Example 2

Solve $x^2y - x^3 \frac{dy}{dx} = y^4 \cos x$

Solution

Re-write: $-x^3 \frac{dy}{dx} + x^2y = y^4 \cos x$

$$\frac{dy}{dx} - \frac{1}{x}y = y^4 \frac{\cos x}{x^3} \quad (\text{standard Bernoulli eqn})$$

Step 1: $(xy^{-4}) \quad y^{-4} \frac{dy}{dx} - \frac{1}{x}y^{-3} = \frac{\cos x}{x^3}$

Step 2: $(x^{-3}) \quad -3y^{-4} \frac{dy}{dx} + \frac{3}{x}y^{-3} = -3 \frac{\cos x}{x^3}$

Step 4: Let $z = y^{-3} \xrightarrow{\frac{d}{dx}} \frac{dz}{dx} = -3y^{-4} \frac{dy}{dx}$

Substitute: $\frac{dz}{dx} + \frac{3}{x}z = -3 \frac{\cos x}{x^3}$ ← (Linear in z and x)

Let $p(x) = \frac{3}{x}$; $\int p(x) dx = \int \frac{3}{x} dx = 3 \ln x = \ln(x^3)$ -3-

$$I.F = e^{\ln(x^3)} = \underline{\underline{x^3}}$$

$$x^3 \left(\frac{dz}{dx} + \frac{3}{x} z \right) = -3 \frac{\cos x}{x^3} \cdot x^3$$

$$\frac{d}{dx} (z \cdot x^3) = -3 \cos x$$

$$z \cdot x^3 = -3 \int \cos x dx$$

$$z \cdot x^3 = -3 \sin x + C$$

$$y^3 = z = -3 \frac{\sin x}{x^3} + \frac{C}{x^3}$$

$$\underline{\underline{y = \left(-\frac{3 \sin x}{x^3} + \frac{C}{x^3} \right)^{-\frac{1}{3}}}}$$

Example 3

Solve: $y - 2x \frac{dy}{dx} = x(x+1)y^3$

Solution

Re-write: $-2x \frac{dy}{dx} + y = x(x+1)y^3$

$$\frac{dy}{dx} - \frac{1}{2x} y = -\frac{1}{2} (x+1) y^3$$

Std
Bernoulli eqn

Step 1:
[x y⁻³

$$\bar{y}^3 \frac{d\bar{y}}{dx} - \frac{1}{2x} \bar{y}^2 = -\frac{1}{2} (x+1)$$

Step 2:
[x-2

$$\underline{\underline{-2\bar{y}^3 \frac{d\bar{y}}{dx} + \frac{1}{x} \bar{y}^2 = x+1}}$$

Step 3:

Let $z = \bar{y}^{-2} \xrightarrow{\frac{d}{dx}} \frac{dz}{dx} = -2\bar{y}^3 \frac{d\bar{y}}{dx}$

$$\frac{dz}{dx} + \frac{1}{x} z = x+1$$

Linear
O.D.E

Apply I.F method

-4-

$$P(x) = \frac{1}{x}; \int P(x) dx = \int \frac{1}{x} dx = \ln x$$

$$IF = e^{\ln x} = \underline{x}$$

$$x \left(\frac{dz}{dx} + \frac{1}{x} z \right) = x(x+1) = x^2 + x$$

$$\frac{d}{dx} (z \cdot x) = x^2 + x$$

$$z \cdot x = \int (x^2 + x) dx = \frac{1}{3} x^3 + \frac{1}{2} x^2 + C$$

$$z = \frac{1}{3} x^2 + \frac{1}{2} x + \frac{C}{x}$$

$$y^{-2} = z = \frac{2x^3 + 3x^2 + 6C}{6x}$$

$$y = \left(\frac{2x^3 + 3x^2 + 6C}{6x} \right)^{-1/2}$$

Put: $A = 6C$

$$y = \left(\frac{6x}{2x^3 + 3x^2 + A} \right)^{1/2}$$

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